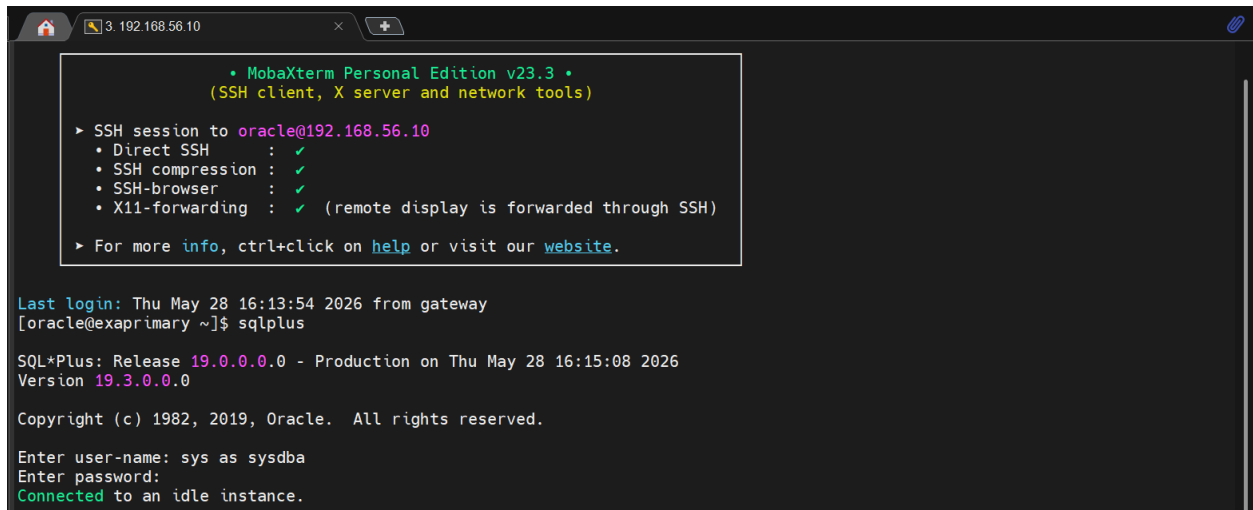


BACKUP 1 SERVER

1. Pertama, masuk ke sqlplus seperti difoto ini



```
• MobaXterm Personal Edition v23.3 •  
(SSH client, X server and network tools)  
  
► SSH session to oracle@192.168.56.10  
• Direct SSH : ✓  
• SSH compression : ✓  
• SSH-browser : ✓  
• X11-forwarding : ✓ (remote display is forwarded through SSH)  
  
► For more info, ctrl+click on help or visit our website.  
  
Last login: Thu May 28 16:13:54 2026 from gateway  
[oracle@exaprimary ~]$ sqlplus  
  
SQL*Plus: Release 19.0.0.0.0 - Production on Thu May 28 16:15:08 2026  
Version 19.3.0.0.0  
  
Copyright (c) 1982, 2019, Oracle. All rights reserved.  
  
Enter user-name: sys as sysdba  
Enter password:  
Connected to an idle instance.
```

2. Menyalakan SQL nya dengan

- startup

3. Cek status Archive log nya

- select log_mode from v\$database;

Jika belum nyala, maka kita nyalakan archive log nya terlebih dahulu

4. Matikan dan nyalakan sql nya menjadi mount

- shut immediate;

- startup mount;

5. Lalu tinggal nyalakan archive lognya

- alter database archive log;

- alter database open;

- archive log list;

Hasilnya akan seperti ini juga sudah nyala:

```
SQL> archive log list;  
Database log mode           Archive Mode  
Automatic archival          Enabled  
Archive destination          USE_DB_RECOVERY_FILE_DEST  
Oldest online log sequence   4  
Next log sequence to archive 6  
Current log sequence         6
```

6. Lalu kita akan membackup Database beserta Archivelog kita, agar suatu hari nanti jika terjadi apa-apa kita tinggal restore, sudah ada backup-an nya

Keluar dari sql

- exit;

Masuk ke rman

- rman target /
- backup database plus archivelog;
- list backup;

jika berhasil maka,

Hasil: RMAN berhasil melakukan backup ke direktori Fast Recovery Area (FRA) yaitu /u01/app/oracle/fast_recovery_area/ORCL/backupset/.

- exit;

Jadi kita sudah berhasil membackup database beserta archivelog di sever kita.

=== Simulasi Skenario ===

A. Simulasi Kerusakan dan Restore Datafile

Skenario ini menyimulasikan hilangnya datafile 7 (users tablespace) secara tidak sengaja di tingkat OS (menggunakan command rm), sehingga menyebabkan database crash.

1. Penghapusan File (Simulasi Error):

- rm /u01/app/oracle/oradata/ORCL/datafile/o1_mf_users_nwbh9cg0_.dbf

2. Identifikasi Kerusakan:

Saat dicoba dimatikan dengan shut immediate, muncul error ORA-01116 dan ORA-01110.

```
[oracle@exaprimary ~]$ sqlplus / as sysdba
SQL*Plus: Release 19.0.0.0.0 - Production on Thu May 28 16:45:41 2026
Version 19.3.0.0.0
Copyright (c) 1982, 2019, Oracle. All rights reserved.

Connected to:
Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Version 19.3.0.0.0

SQL> shut immediate;
ORA-01116: error in opening database file 7
ORA-01110: data file 7: '/u01/app/oracle/oradata/ORCL/datafile/o1_mf_users_o1g80xs5_.dbf'
ORA-27041: unable to open file
Linux-x86_64 Error: 2: No such file or directory
Additional information: 3
```

Database kemudian di-shutdown paksa dengan shut abort dan dihidupkan sampai tahap mount.

- shut abort;
- startup mount;

3. Proses Restore dan Recover:

- rman target /
- restore datafile 7;
- recover datafile 7;
- alter database open;

```
RMAN> restore datafile 7;

Starting restore at 28-MAY-26
using target database control file instead of recovery catalog
allocated channel: ORA_DISK_1
channel ORA_DISK_1: SID=433 device type=DISK

channel ORA_DISK_1: starting datafile backup set restore
channel ORA_DISK_1: specifying datafile(s) to restore from backup set
channel ORA_DISK_1: restoring datafile 00007 to /u01/app/oracle/oradata/ORCL/datafile/o1_mf_users_o1g80xs5_.dbf
channel ORA_DISK_1: reading from backup piece /u01/app/oracle/fast_recovery_area/ORCL/backupset/2026_05_28/o1_mf_nnndf_TAG2026
0528T162755_o1j2nwcx_.bkp
channel ORA_DISK_1: piece handle=/u01/app/oracle/fast_recovery_area/ORCL/backupset/2026_05_28/o1_mf_nnndf_TAG20260528T162755_o
1j2nwcx_.bkp tag=TAG20260528T162755
channel ORA_DISK_1: restored backup piece 1
channel ORA_DISK_1: restore complete, elapsed time: 00:00:01
Finished restore at 28-MAY-26

RMAN> recover datafile 7;

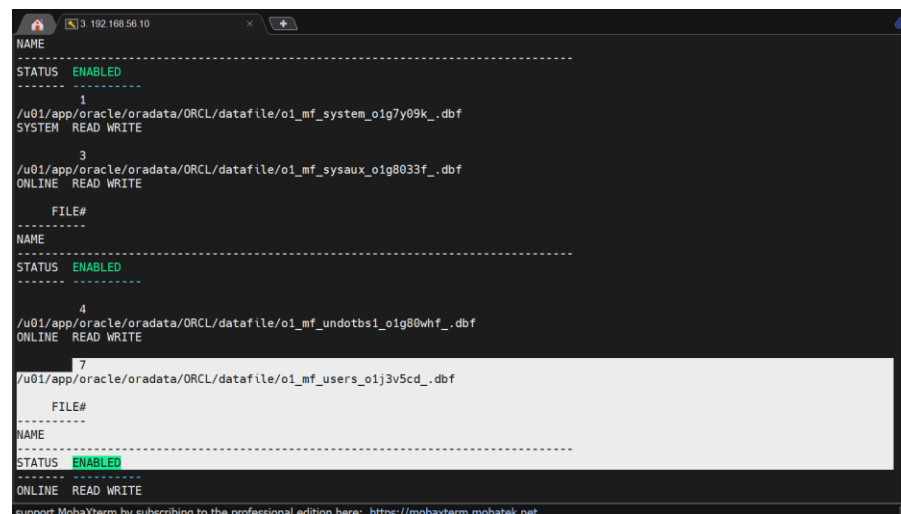
Starting recover at 28-MAY-26
using channel ORA_DISK_1

starting media recovery
media recovery complete, elapsed time: 00:00:01
Finished recover at 28-MAY-26
```

Masuk ke sql untuk mengecek apakah sudah ada filenya atau belum

- DESC v\$datafile;
- SELECT file#, name, status, enabled FROM v\$datafile;

Hasil: Datafile 7 berhasil di-restore dan di-recover sepenuhnya. Database kembali dalam status OPEN.



```
NAME
-----
STATUS  ENABLED
-----
1
/u01/app/oracle/oradata/ORCL/datafile/o1_mf_system_o1g7y99k_.dbf
SYSTEM  READ WRITE

3
/u01/app/oracle/oradata/ORCL/datafile/o1_mf_sysaux_o1g8033f_.dbf
ONLINE  READ WRITE

FILE#
-----
NAME
-----
STATUS  ENABLED
-----
4
/u01/app/oracle/oradata/ORCL/datafile/o1_mf_undots1_o1g80whf_.dbf
ONLINE  READ WRITE

7
/u01/app/oracle/oradata/ORCL/datafile/o1_mf_users_o1j3v5cd_.dbf
FILE#
-----
NAME
-----
STATUS  ENABLED
-----
ONLINE  READ WRITE

support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net
```

B. Logical Backup dengan Data Pump (expdp & impdp)

Proses ini bertujuan untuk mengekspor skema C##ADMIN_TUGAS dan mengimpornya kembali. Sempat terjadi kendala pada Listener dan User Role.

1. Membuat Directory Objek

- mkdir -p /u01/app/oracle/admin/ORCL/dpdump

2. Masuk ke Sql untuk membuat direktori dan member akses direktori tersebut ke sistem

- create or replace directory my_backup_dir as '/u01/app/oracle/admin/ORCL/dpdump';
- GRANT READ, WRITE ON DIRECTORY my_backup_dir TO system;

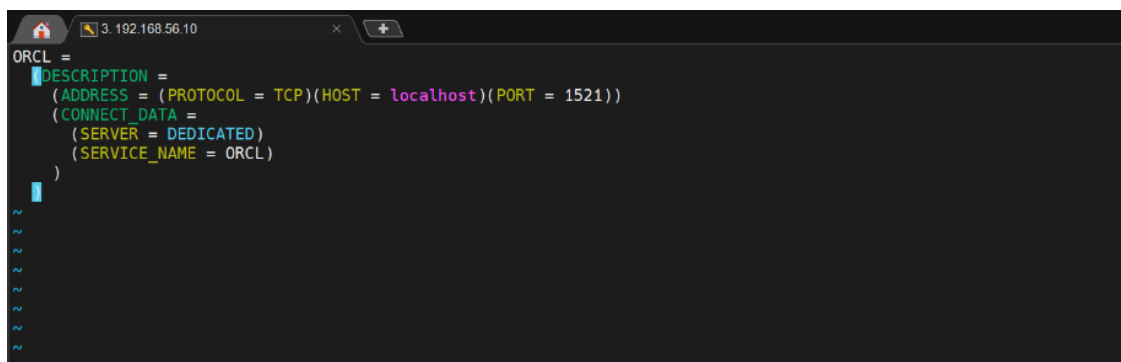
3. Lalu keluar dari sql, dan nyalakan listener controlnya

- lsnrctl start

4. Kita akan membuat tnsnames untuk dapat mengecek ping, terhubung dengan listener atau tidak dengan tnsping

- vi \$ORACLE_HOME/network/admin/tnsnames.ora

=== SCRIPT ===



```
ORCL =  
  (DESCRIPTION =  
    (ADDRESS = (PROTOCOL = TCP)(HOST = localhost)(PORT = 1521))  
    (CONNECT_DATA =  
      (SERVER = DEDICATED)  
      (SERVICE_NAME = ORCL)  
    )  
  )  
~  
~  
~  
~  
~  
~  
~
```

Lalu Cek ping nya

- tnsping ORCL

Jika Hasilnya berhasil terhubung maka kita sudah siap untuk ekspor dan impor.

5. Membuat User dan Ekspor data

Karena skema C##ADMIN_TUGAS belum ada, dilakukan pembuatan user terlebih dahulu sebelum diekspor.

Masuk ke sql lalu:

- ALTER SESSION SET "_ORACLE_SCRIPT"=TRUE;
- CREATE USER "C##ADMIN_TUGAS" IDENTIFIED BY "123";
- GRANT CONNECT, RESOURCE, DBA TO "C##ADMIN_TUGAS";
- ALTER USER C##ADMIN_TUGAS QUOTA UNLIMITED ON USERS;
- exit;

Lalu tinggal export skema C##ADMIN_TUGAS ke direktori yang tadi kita buat

- expdp system/123@ORCL directory=my_backup_dir dumpfile=backup_belanjatugas.dmp logfile=export_log.log schemas=C##ADMIN_TUGAS

```
[oracle@exapriary ~]$ expdp system/123@ORCL directory=my_backup_dir dumpfile=backup_belanjatugas.dmp logfile=export_log.log s
chemas=C##ADMIN_TUGAS

Export: Release 19.0.0.0.0 - Production on Thu May 28 17:16:04 2026
Version 19.3.0.0.0

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Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Starting "SYSTEM"."SYS_EXPORT_SCHEMA_01": system/*****@ORCL directory=my_backup_dir dumpfile=backup_belanjatugas.dmp logfi
le=export_log.log schemas=C##ADMIN_TUGAS
Processing object type SCHEMA_EXPORT/SYSTEM_GRANT
Processing object type SCHEMA_EXPORT/ROLE_GRANT
Processing object type SCHEMA_EXPORT/DEFAULT_ROLE
Processing object type SCHEMA_EXPORT/TABLESPACE_QUOTA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA
Master table "SYSTEM"."SYS_EXPORT_SCHEMA_01" successfully loaded/unloaded
*****
Dump file set for SYSTEM.SYS_EXPORT_SCHEMA_01 is:
/u01/app/oracle/admin/ORCL/dpdump/backup_belanjatugas.dmp
Job "SYSTEM"."SYS_EXPORT_SCHEMA_01" successfully completed at Thu May 28 17:17:30 2026 elapsed 0 00:01:09

[oracle@exapriary ~]$
```

Catatan: Jika sudah seperti ini Ekspor berhasil, namun karena skema belum memiliki tabel, maka yang terekspor hanya metadata (role, grants, quota).

6. Disini kita akan sengaja menghapus skema nya (Simulasi jika suatu saat tidak sengaja menghapus sesuatu)

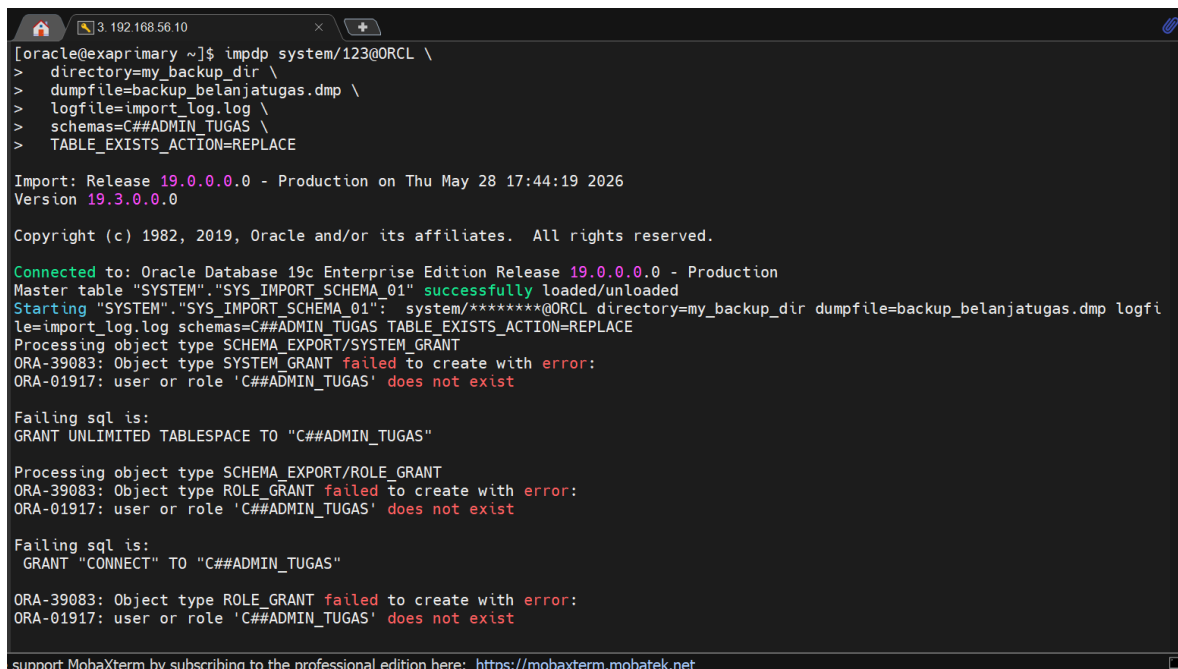
- ALTER SESSION SET "_ORACLE_SCRIPT"=TRUE;
- DROP USER C##ADMIN_TUGAS CASCADE;

Verifikasi untuk memastikan user sudah hilang:

- SELECT username FROM dba_users WHERE username = 'C##ADMIN_TUGAS';

Hasil: no rows selected

7. Jika kita langsung Import maka akan mengalami error seperti ini:



```
[oracle@exapriary ~]$ impdp system/123@ORCL \
> directory=my_backup_dir \
> dumpfile=backup_belanjatugas.dmp \
> logfile=import_log.log \
> schemas=C##ADMIN_TUGAS \
> TABLE_EXISTS_ACTION=REPLACE

Import: Release 19.0.0.0.0 - Production on Thu May 28 17:44:19 2026
Version 19.3.0.0.0

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Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Master table "SYSTEM"."SYS_IMPORT_SCHEMA_01" successfully loaded/unloaded
Starting "SYSTEM"."SYS_IMPORT_SCHEMA_01": system/*****@ORCL directory=my_backup_dir dumpfile=backup_belanjatugas.dmp logfi
le=import_log.log schemas=C##ADMIN_TUGAS TABLE_EXISTS_ACTION=REPLACE
Processing object type SCHEMA_EXPORT/SYSTEM_GRANT
ORA-39083: Object type SYSTEM_GRANT failed to create with error:
ORA-01917: user or role 'C##ADMIN_TUGAS' does not exist

Failing sql is:
GRANT UNLIMITED TABLESPACE TO "C##ADMIN_TUGAS"

Processing object type SCHEMA_EXPORT/ROLE_GRANT
ORA-39083: Object type ROLE_GRANT failed to create with error:
ORA-01917: user or role 'C##ADMIN_TUGAS' does not exist

Failing sql is:
GRANT "CONNECT" TO "C##ADMIN_TUGAS"

ORA-39083: Object type ROLE_GRANT failed to create with error:
ORA-01917: user or role 'C##ADMIN_TUGAS' does not exist
```

- **Alasan Error ORA-01917:** Sebenarnya, impdp bisa membuat user secara otomatis jika kita menggunakan user biasa (Local User). Namun, karena kamu menggunakan Common User (C##ADMIN_TUGAS) di Oracle 19c (arsitektur Container/Multitenant), Oracle memberikan proteksi keamanan yang sangat ketat. Di Oracle 19c, membuat user beratribut C## memerlukan parameter khusus di latar belakang yaitu `_ORACLE_SCRIPT=TRUE`.

Karena utilitas impdp berjalan sebagai background job otomatis, dia tidak bisa mengaktifkan parameter tersebut sendiri saat mencoba mengeksekusi perintah `CREATE USER "C##ADMIN_TUGAS"`, sehingga prosesnya digagalkan oleh sistem Oracle dan memicu error ORA-01917.

8. Solusinya kita buat ulang struktur dasar user (Infrastruktur/Metadata) di SQL terlebih dahulu:

- `ALTER SESSION SET "_ORACLE_SCRIPT"=TRUE;`
- `CREATE USER C##ADMIN_TUGAS IDENTIFIED BY 123;`
- `GRANT CONNECT, RESOURCE, DBA TO C##ADMIN_TUGAS;`
- `ALTER USER C##ADMIN_TUGAS QUOTA UNLIMITED ON USERS;`

9. Setelah user dasar dibuat kembali, jalankan perintah import di **Terminal Linux** (bukan di SQL*Plus):

- `impdp system/123@ORCL directory=my_backup_dir dumpfile=backup_belanjatugas.dmp logfile=import_log.log schemas=C##ADMIN_TUGAS TABLE_EXISTS_ACTION=REPLACE`

```

[oracle@exaprimary ~]$ impdp system/123@ORCL directory=my_backup_dir dumpfile=backup_belanjatugas.dmp logfile=import_log.log schemas=C##ADMIN_TUGAS TABLE_EXISTS_ACTION=REPLACE

Import: Release 19.0.0.0.0 - Production on Thu May 28 17:58:30 2026
Version 19.3.0.0.0

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Connected to: Oracle Database 19c Enterprise Edition Release 19.0.0.0.0 - Production
Master table "SYSTEM"."SYS_IMPORT_SCHEMA_01" successfully loaded/unloaded
Starting "SYSTEM"."SYS_IMPORT_SCHEMA_01": system/*****@ORCL directory=my_backup_dir dumpfile=backup_belanjatugas.dmp logfile=import_log.log schemas=C##ADMIN_TUGAS TABLE_EXISTS_ACTION=REPLACE
Processing object type SCHEMA_EXPORT/SYSTEM_GRANT
Processing object type SCHEMA_EXPORT/ROLE_GRANT
Processing object type SCHEMA_EXPORT/DEFAULT_ROLE
Processing object type SCHEMA_EXPORT/TABLESPACE_QUOTA
Processing object type SCHEMA_EXPORT/PRE_SCHEMA/PROCACT_SCHEMA
Job "SYSTEM"."SYS_IMPORT_SCHEMA_01" successfully completed at Thu May 28 17:58:39 2026 elapsed 0 00:00:05

[oracle@exaprimary ~]$

```

Jadi, Kita sudah berhasil meng-Import file yang tidak sengaja didrop tadi dengan impdp.

Lalu pasti bingung, “Kalau kita harus membuat user-nya lagi secara manual, lalu apa gunanya impdp?”

Jawabannya adalah: impdp bukan sekadar untuk membuat nama user, melainkan untuk mengembalikan seluruh "isi" dan aset di dalam skema tersebut.

Dibuat oleh Naufal Ali Hilmi | Information Systems Student, Gunadarma University